Lab 3: Component Modelling & Architectural Pattern Selection

Written Justification

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#53 Problem statement : Freelance Content Creator Escrow Platform

Architecture Selection : We chose **Layered Architecture** for the Freelance Content Creator Escrow Platform. The architecture separates the system into Presentation, Business, and Data layers, providing a clear organization of responsibilities.

Reason 1 – Clear separation of responsibilities : The Presentation Layer handles user interaction, the Business Layer manages project and contract operations, content management, and escrow/payment processing, and the Data Layer manages persistent platform data. This separation makes the system easier to understand, develop, and maintain.

Reason 2 – Suitable for the platform workflow : The platform contains distinct activities such as project and contract management, content submission, escrow/payment processing, and data storage. Layered Architecture organizes these responsibilities into appropriate layers and allows them to communicate through defined interfaces.

Security Advantage: Layered Architecture helps protect sensitive escrow and payment information by keeping data access within the Data Layer and applying business rules in the Business Layer. The user interface therefore does not directly access the database.

Performance Benefit : Separating database operations from presentation and business logic allows data access to be optimized independently and avoids unnecessary direct database interactions from the user interface.

Conclusion : Layered Architecture is appropriate for the platform because it provides modularity, separation of concerns, maintainability, controlled access to data, and organized communication between system components.